

Computer Organization and Assembly Language (EL-2003)

Date: Monday, October 06, 2025

Time: 11:00 AM – 12:40 PM

Instructor(s)

Waseem Rauf

Lab Mid-Exam

Total Time: 100 minutes

Total Marks: 20

Total Questions: 03

Semester: Fall 2025

Paper: B

Student Name

Roll No

Section

Student Signature

write below this line

02: Understanding and apply assembly language programming concepts, including data definition, initialization, and manipulation, as well as the use of instructions, loops, and built-in procedures to perform complex tasks, and demonstrate an understanding of how to work with arrays, conditional processing, strings, and flags to control program flow and perform input/output operations

Question # 01

[Marks: 05 | Time: 20 Minutes]

i) Write down the value of AX.

data

byteValue BYTE 11100011b

main proc

movsx ax, byteValue

ii) Given the code, what is the value returned by SIZEOF arrayB, and why is the result four times greater than the value returned by LENGTHOF arrayB?

data

arrayB DWORD 4 DUP(10)

iii) Given the Assembly Language code snippet below, determine the full 6-byte sequence of values stored in memory for the variable data_set. Based on this sequence, what is the final decimal value loaded into the AL register by the MOV instruction?

data

data_set BYTE 3 DUP(5, 15)

code

main PROC

MOV al, data_set + 4

iv) Difference between Carry Flag and Overflow Flag with example.

National University of Computer and Emerging Sciences

Karachi Campus

v) A single MOV instruction cannot be used to move data directly from one memory location to another. Write down a series of instructions to move data from one word variable to other.

LLO # 02: Understanding and apply assembly language programming concepts, including data definition, initialization, and manipulation, as well as the use of instructions, loops, and built-in procedures to perform complex tasks, and demonstrate an understanding of how to work with arrays, conditional processing, strings, and flags to control program flow and perform input/output operations

Question # 02

[Marks: 4+4 = 08 | Time: 45 Minutes]

(i) Write a program with a loop and indexed addressing that calculates the sum of all the gaps between successive array elements. The array elements are words, sequenced in nondecreasing order. So, for example, the array {0, 2, 5, 9, 10} has gaps of 2, 3, 4, and 1, whose sum equals 10

(ii) Write an assembly program with a loop and indexed addressing that exchanges every pair of values in an array with an even number of elements. Therefore, item i need to exchange with item $i + 2$, and item $i + 1$ will exchange with item $i + 3$, and so on. Example: [1,2,3,4,5,6,7,8] $\rightarrow$ [3,4,1,2,7,8,5,6]. Input should be taken from user.

LLO # 02: Understanding and apply assembly language programming concepts, including data definition, initialization, and manipulation, as well as the use of instructions, loops, and built-in procedures to perform complex tasks, and demonstrate an understanding of how to work with arrays, conditional processing, strings, and flags to control program flow and perform input/output operations

Question # 03

[Marks: 4+3 = 07 | Time: 35 Minutes]

(i) Implement this nested loop into assembly language and print the final total_sum value:

```
int total_sum = 0;
int number = 5 ;
for (int i = 0; i < 5 ; i++) {
    for (int j = 0; j <= i; j++) {
        total_sum += number ;
    }
}
```

(ii) Solve the Following Mathematics Expression by Using Registers:

a) $(9 - 4) + (9 + 7)$

b) $x = 9 + w + (--x) + (y++) - z$ here $w=4$; $x=5$; $y=7$; $z=10$